ADARSH GORREMUCHU

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PROFESSIONAL SUMMARY

Senior Data Scientist with an M.S. in Data Analytics Engineering and 4+ years of experience in statistical programming, predictive modeling, machine learning, and data storytelling across complex datasets. Strong proficiency in Python, SQL, PySpark, and SparkSQL on Azure Databricks for large-scale exploratory data analysis, model development, and lifecycle execution. Experienced building AI/ML models with fairness, governance, and responsible AI considerations; skilled at applying disparate impact analysis and bias assessment techniques aligned with fair lending and consumer compliance best practices. Adept at translating statistical insights into clear, compelling narratives for technical and non-technical stakeholders through Power BI, Tableau, and data visualization dashboards.

EXPERIENCE

AI/ML Engineer & Data Scientist Jun 2025 – Present

Ampcus Inc Chantilly, VA

- Designed and evaluated predictive AI/ML models end-to-end — from feature engineering and hypothesis testing to modelvalidation and lifecycle documentation — ensuring responsible AI and ethical use of data aligned with governance standards.
- Applied disparate impact analysis and statistical bias assessment techniques to review model outputs and business policiesfor potential fairness risks; collaborated with compliance and legal stakeholders to support risk mitigation decisions.
- Conducted descriptive, predictive, and prescriptive analyses on large, complex datasets using Python and PySpark on AzureDatabricks; extracted actionable insights from member and transactional data to drive mission-critical decisions.
- Built automated analytical processes and moderately complex algorithms in Python for classification, summarization, andanomaly detection; reduced analysis turnaround by 35% and supported ongoing and ad hoc compliance review workflows.
- Developed data visualizations, dashboards, and executive presentations using Power BI to communicate analytical findings,model results, and risk monitoring insights to both technical teams and non-technical leadership.
- Wrote clear technical documentation and data storytelling narratives to convey complex statistical concepts to diverseaudiences including business partners, compliance, and legal subject-matter experts.

Cloud Data Engineer & ML Developer May 2024 – Nov 2024

Community Informatics Lab (George Mason University) Fairfax, VA

- Developed and evaluated machine learning pipelines using Python, PyTorch, and TensorFlow; applied statistical methods — regression, correlation, variance, standard deviation, clustering — to test hypotheses and improve model performance on research benchmarks by 35%.
- Performed exploratory data analysis (EDA) on large, complex datasets; identified meaningful patterns, tested modelassumptions, and documented analytical results to support data science and business intelligence projects.
- Built PySpark and SparkSQL pipelines on Databricks for large-scale data ingestion, transformation, and modeling simulation;accelerated runtimes by 55% while sustaining 99.9% data reliability and quality standards.
- Created Power BI reports and data visualizations to support research analysis and stakeholder decision-making; applied datastorytelling techniques to communicate statistical insights across technical and non-technical audiences.
- Designed multi-cloud data architecture (Snowflake, BigQuery, NoSQL) with standardized schemas, data quality monitoring,and governed access paths supporting 20+ analytical projects.

Data Engineer & BI Developer Dec 2020 – Dec 2022

TriSX Global India Pvt Ltd Hyderabad, India

- Applied predictive modeling and statistical analysis using Python and TensorFlow; performed feature engineering,cross-validation, and trend analysis to improve KPI forecasting accuracy by 20% and support business planning decisions.

- Developed and coded automated analytical processes and ETL pipelines on Databricks, Spark, and cloud warehouses(Redshift, Snowflake, BigQuery); reduced reporting latency by 55% and improved data integrity monitoring.
- Built executive Power BI dashboards with standardized KPI definitions and data visualizations; improved stakeholderadoption and reporting consistency by 30% across cross-functional teams through clear data storytelling.
- Resolved complex data quality issues using advanced profiling, cleansing, and monitoring techniques; maintained dataintegrity and reliability across warehouse ingestion and reporting workflows.

KEY PROJECTS

Bias & Fairness Assessment – AI Model Governance *Python · PySpark · Azure Databricks · Statistical Analysis*

- Reviewed AI/ML model inputs, assumptions, and outcomes for potential fairness risk; applied disparate impact analysis andstatistical testing to evaluate model outputs against compliance and responsible AI governance frameworks.
- Documented analytical findings and model review results in technical writing reports, collaborating with stakeholders tosupport risk mitigation and corrective action planning.

Predictive Ranking & Campaign Outcome Model *Python · Machine Learning · AWS · Power BI · SQL*

- Designed, developed, and deployed a predictive ML model with optimized feature selection to forecast campaign outcomes;integrated results into Power BI dashboards enabling data-driven decisions for 10+ stakeholders.

Serverless Semantic Search & Analytics Pipeline *Python · OpenSearch · PySpark · Azure Databricks · Terraform*

- Built scalable analytical pipelines for classification, data summarization, and context extraction; achieved sub-secondretrieval latency through query optimization and structured dataset design on Azure Databricks.

TECHNICAL SKILLS

Statistical Programming: Python, SQL, PySpark, SparkSQL, R, SAS/Stat, Bash

AI/ML Models: Predictive Modeling, Regression, Clustering, Classification, NLP, LLMs, Model Lifecycle, Modeling Simulations

Fair Lending & Compliance: Disparate Impact Analysis, Bias Assessment, Responsible AI, Ethical AI, Model Governance, Fair Lending Frameworks

Data Engineering: Azure Databricks, Spark, PySpark, SparkSQL, Azure Data Factory, Azure Data Lake, Snowflake, BigQuery, Redshift

Data Visualization: Power BI, Tableau, Data Storytelling, Dashboard Development, Creating Presentations

Statistics & Analytics: Regression, Correlation, Variance, Standard Deviation, Normal Distribution, Hypothesis Testing, Trend Analysis, Insights

Cloud & MLOps: AWS (Lambda, S3, Step Functions), Azure AI, Docker, Terraform, Airflow

Communication: Technical Writing, Written and Verbal Communication, Data Storytelling, Stakeholder Presentations

EDUCATION

M.S., Data Analytics Engineering Jan 2023 – Dec 2024

George Mason University, Fairfax, VA | Focus: Machine Learning, Statistics, Data Engineering, Cloud Computing